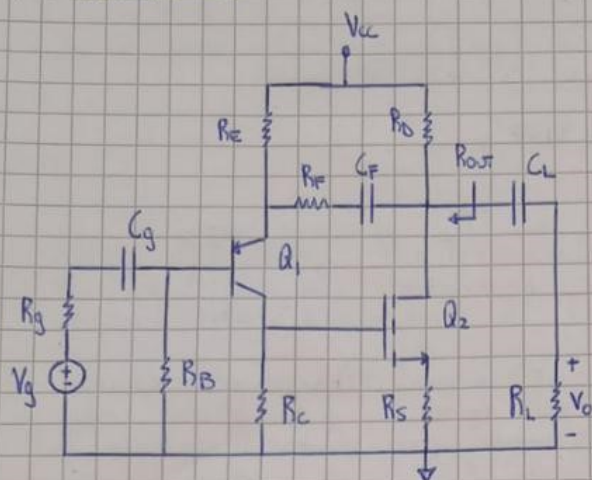


ESERCIZIO 1



Consider the two-stage amplifier with the parameters at 25°C :

$$\begin{aligned} V_{CC} &= 15 \text{ V} \\ R_B &= 1,8 \text{ M}\Omega \\ R_E &= 1,2 \text{ k}\Omega \\ R_S &= 1 \text{ k}\Omega \\ R_F &= 2,2 \text{ k}\Omega \\ C_L &= 1 \mu\text{F} \end{aligned}$$

$$\begin{aligned} R_g &= 10 \text{ k}\Omega \\ R_C &= 10 \text{ k}\Omega \\ R_D &= 6,8 \text{ k}\Omega \\ R_L &= 100 \text{ k}\Omega \\ C_g &= 10 \text{ nF} \\ C_F &= 1 \mu\text{F} \end{aligned}$$

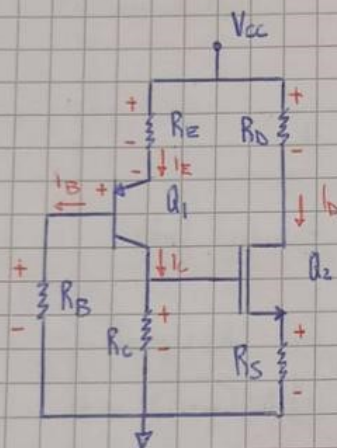
$$Q_1: V_{BE} = -0,7 \text{ V} \\ V_0 = +\infty$$

$$\beta_F = \beta_0 = 100 \\ V_T = 25 \text{ mV}$$

$$Q_2: V_{T0} = 2 \text{ V} \\ V_0 = +\infty$$

$$I_{DSS} = K_n \frac{W}{L} V_T^2 = 0,2 \text{ mA}$$

- 1) Considering all capacitors to be equivalent to open circuits, determine the operating points (V_{CE}, I_C) of Q_1 and (V_{GS}, I_D) of Q_2 .



Multistadio: $CE + RE$ e $CS + RS$

NB: C_g, C_F e C_L condensatori di accoppiamento

$$V_{CC} - R_E (\beta_F + 1) I_B + V_{BE} - R_B I_B = 0$$

$$\rightarrow I_B = \frac{V_{CC} + V_{BE}}{(\beta_F + 1) R_E + R_B} = 7,44 \mu\text{A} > 0$$

$$I_C = \beta_F I_B = 0,74 \text{ mA} \quad I_E = (\beta_F + 1) I_B = 0,75 \text{ mA}$$

$$V_{CE} = (R_C I_C) - (V_{CC} - R_E I_E) = -6,7 \text{ V} < V_{CE, \text{SAT}} \quad Q_1 \text{ in FAB}$$

Per n-mos: $I_D = I_{DSS} \left(\frac{V_{GS}}{V_T} - 1 \right)^2$ e $V_{GS} = R_C I_C - R_S I_D$

Risolve per V_{GS} perché è immediato distinguere il valore buono:

$$V_{GS} = R_C I_C - R_S I_{DSS} \left(\frac{V_{GS}}{V_T} - 1 \right)^2 \quad V_{GS} = R_C I_C - R_S I_{DSS} \left(\frac{V_{GS}^2}{V_T^2} - 2 \frac{V_{GS}}{V_T} + 1 \right)$$

$$V_{GS} = R_C I_C - \frac{R_S I_{DSS}}{V_T^2} V_{GS}^2 + \frac{2 R_S I_{DSS}}{V_T} V_{GS} - R_S I_{DSS}$$

$$V_{GS}^2 \left(\frac{R_S I_{DSS}}{V_T^2} \right) + V_{GS} \left(1 - \frac{2 R_S I_{DSS}}{V_T} \right) + (-R_C I_C + R_S I_{DSS}) = 0$$

$a = 0,05 \quad b = 0,8 \quad c = -7,2$

$$V_{GS} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-0,8 \pm 1,44}{0,1} = \begin{cases} 6,45 \text{ V} > V_T & \text{OK} \\ -22,4 \text{ V} < V_T & \text{NO} \end{cases}$$

$$I_D = I_{DSS} \left(\frac{V_{GS}}{V_T} - 1 \right)^2 = 0,98 \text{ mA}$$

$$V_{DS} = (V_{CC} - R_D I_D) - (I_D R_S) = 7,4 \text{ V} > V_{GS} - V_T = 4,45 \text{ V} \quad \text{OK } Q_2 \text{ in SAT}$$

The bias points are: $Q_1 (-6,7 \text{ V}, 0,74 \text{ mA})$ e $Q_2 (7,4 \text{ V}, 0,98 \text{ mA})$ ④

2) Assuming all capacitors to be equivalent to short circuits at the frequency of interest, determine:

- The voltage gain $A_v = v_o/v_g$ of the amplifier
- The output resistance R_{out} indicated in the figure
- The low frequency bandwidth limit of the amplifier

$$g_{m1} = 40 I_{C1} = 29,6 \text{ mS}$$

$$r_{\pi1} = R_o / g_{m1} = 3378 \Omega$$

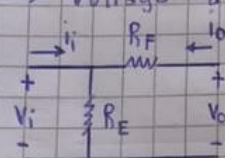
$$r_{o1} \rightarrow +\infty$$

$$g_{m2} = \frac{2}{V_T} \sqrt{I_{D1} I_{DSS}} = 0,44 \text{ mS}$$

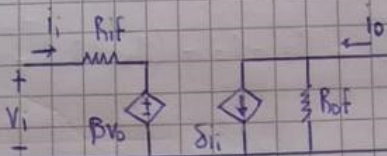
$$r_{o2} \rightarrow +\infty$$

Voltage/series mixing: feedback modifies V_E (control Q_1 varying V_{BE} not I_B)
Voltage/shunt sampling: if I short R_E I kill the feedback (no current coming from the output anymore).

$\Rightarrow$ voltage amplifier



$\Leftrightarrow$



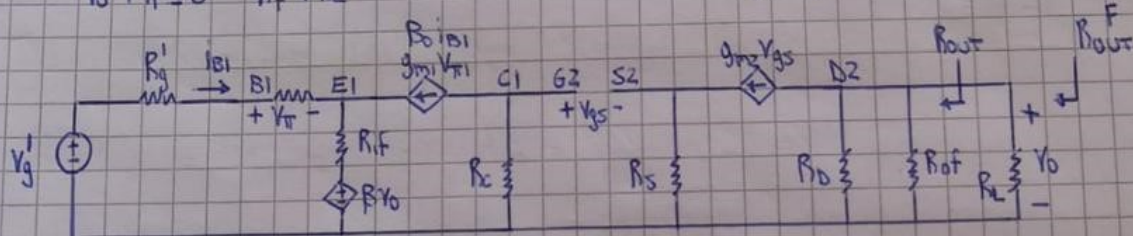
Feedback network must include R_F and R_E (also R_D , it doesn't change) because I need a voltage divider to impose a voltage.

$$R_{if} = \frac{V_i}{i_i} \bigg|_{V_o=0} = R_E \parallel R_F = 776 \Omega$$

$$R_{of} = \frac{V_o}{i_o} \bigg|_{i_i=0} = R_E + R_F = 3400 \Omega$$

$$\beta = \frac{V_i}{V_o} \bigg|_{i_i=0} = \frac{R_E}{R_F + R_E} = 0,353$$

$$\delta = \frac{i_o}{i_i} \bigg|_{V_o=0} = \rightarrow 0 \text{ unilaterality}$$



$$\frac{v_g'}{v_g} = \frac{R_B}{R_B + R_g}$$

$$R_g' = R_B \parallel R_g \approx R_g$$

$$R_L' = R_D \parallel R_{of} \parallel R_L = 2216 \Omega$$

$$A_{OL} = \frac{V_o}{v_g'} \bigg|_{\beta=\delta=0} = \frac{V_{c1}}{v_g'} \cdot \frac{V_o}{V_{c1}} = \frac{-\beta_0 R_C}{R_g' + r_{\pi1} + (\beta_0 + 1) R_{if}} \cdot \frac{-g_{m2} R_L'}{1 + g_{m2} R_S} = -10,9 \cdot (-0,63) = +6,8$$

$$R_{out}^{OL} = R_D \parallel R_{of} \parallel R_L = 2216 \Omega$$

$$A_F = \frac{V_o}{v_g} = \frac{A_{OL}}{1 + \beta A_{OL}} = +2,004 \approx \frac{1}{\beta} = 2,83 \text{ due to low } T = \beta A_{OL} = 2,4$$

$$A_v = \frac{V_o}{v_g} = \frac{V_o}{v_g'} \cdot \frac{v_g'}{v_g} = A_F \cdot \frac{R_B}{R_B + R_g} = +1,99$$

$$R_{out}^F = \frac{R_{out}^{OL}}{1 + \beta A_{OL}} = 650 \Omega$$

$$R_{out} = \frac{1}{1/R_{out}^F - 1/R_L} = 654 \Omega$$

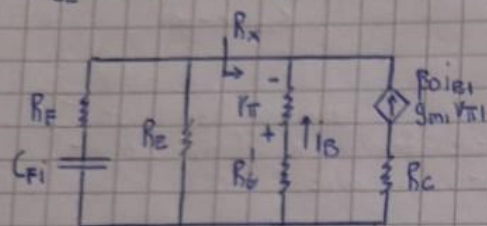
To calculate f_L^{CL} we apply SCTC method to all low frequency capacitors C_L , C_F and C_G ; we have transferred the loading effects of B-network on input and output so we must split C_F in the two sides C_{Fi} and C_{Fo} :

$$f_L^{OL} = \frac{1}{2\pi} \frac{1}{\tau_L} = \frac{1}{2\pi} \left(\frac{1}{C_G R_{Gg}} + \frac{1}{C_L R_{Lc}} + \frac{1}{C_{Fi} R_{CFi}} + \frac{1}{C_{Fo} R_{CFo}} \right)$$

$$R_{Gg}^{SC} = R_G + (R_B \parallel R_{inL}) = 88202 \, \Omega$$

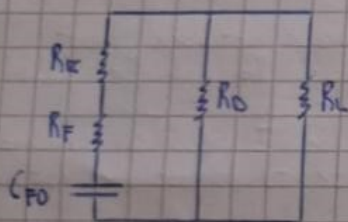
$$R_{inL} = r_{\pi 1} + (\beta_0 + 1) R_{if} = 81754 \, \Omega$$

$$R_{Lc}^{SC} = R_L + (R_{of} \parallel R_D) = 102267 \, \Omega$$



$$R_x = \frac{V_x}{I_x} = \frac{-(R_B' + r_{\pi 1}) I_B}{-(\beta_0 + 1) I_B} = \frac{R_B' + r_{\pi 1}}{(\beta_0 + 1)} = 132 \, \Omega$$

$$R_{CFi}^{SC} = R_F + R_E \parallel R_x = 2319 \, \Omega$$

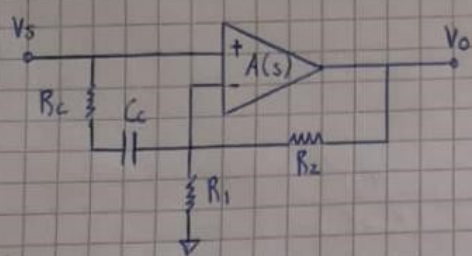


$$R_{CFo}^{SC} = R_F + R_E + (R_D \parallel R_L) = 9767 \, \Omega$$

$$\Rightarrow f_L^{OL} = \frac{1}{2\pi} \frac{1}{\tau_L} = 267 \, \text{Hz}$$

$$f_L^{CL} = \frac{f_L^{OL}}{1 + \beta A_{OL}} = 78,5 \, \text{Hz}$$

ESERCIZIO 2



Consider the operational amplifier with noise gain compensation and:

$$R_1 = 18 \text{ k}\Omega$$

$$R_2 = 56 \text{ k}\Omega$$

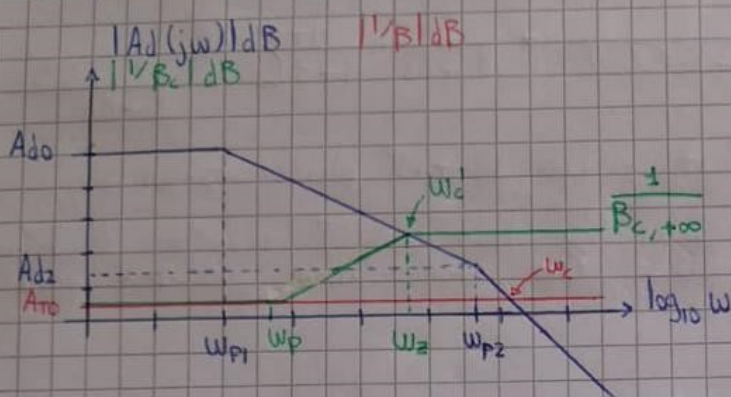
$$A(s) = \frac{A_0}{(1 + s/\omega_{p1})(1 + s/\omega_{p2})} \quad [\text{V/V}]$$

$$A_0 = 10^5 \text{ [V/V]}$$

$$\omega_{p2} = 5 \cdot 10^5 \text{ rad/s}$$

$$\omega_{p1} = 10^2 \text{ rad/s}$$

1) Determine an estimation of circuit phase margin with $R_c \rightarrow +\infty$



$$B = \frac{V_-}{V_o} \bigg|_{V_s=0} = \frac{R_1}{R_1 + R_2}$$

$$A_T = \frac{1}{B} = \frac{R_1 + R_2}{R_1} = 4 \Rightarrow 12 \text{ dB}$$

Closing/opening ratios are $\pm 40 \text{ dB/dec} \rightarrow \text{PM} \rightarrow 0^\circ$

The approximation isn't good because ω_c is near ω_{p2} (the distance is less than a decade); so more precisely

$$A_0 \omega_{p1} = A_{d2} \omega_{p2} \rightarrow A_{d2} = \frac{A_0 \omega_{p1}}{\omega_{p2}} = 20 = 26 \text{ dB}$$

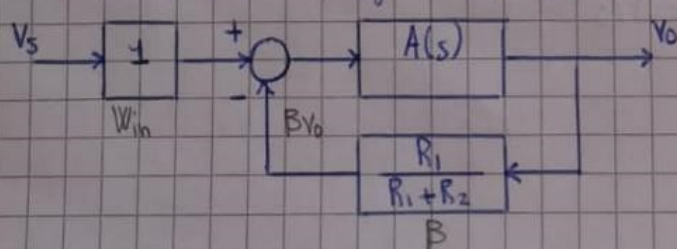
$$A_{d2} \omega_{p2}^2 = A_0 \omega_c^2 \rightarrow \omega_c = \sqrt{\frac{A_{d2} \omega_{p2}^2}{A_0}} = 1.1 \cdot 10^6 \text{ rad/s}$$

$$\text{PM} = 180^\circ + \angle T(j\omega_c) = 180^\circ + \angle \frac{1}{(1 + j\omega_c/\omega_{p1})(1 + j\omega_c/\omega_{p2})} =$$

$$= 180^\circ - \arctg(10000) - \arctg(2.2) = 180^\circ - 90^\circ - 65.5^\circ = 24.4^\circ$$

2) A block diagram representation of the amplifier with $R_c \rightarrow +\infty$, according with feedback theory

It is a non-inverting amplifier so:



NB: it has voltage mixing and voltage sampling.

3) The expression of the closed loop gain of the compensated amplifier, assuming $1 \ll T \ll +\infty$

$$A_F = \frac{A_{OL}}{1 + \beta A_{OL}} \stackrel{T \gg 1}{=} \frac{1}{\beta(s)}$$

$$Z_{cc} = R_c + 1/sC_c = \frac{1 + sC_c R_c}{sC_c}$$

$$R_1 // Z_{cc} = \frac{R_1 (1 + sC_c R_c)}{sC_c} \cdot \frac{1}{1 + sC_c R_c + sC_c R_1} = \frac{R_1 (1 + sC_c R_c)}{1 + sC_c (R_1 + R_c)}$$

$$\begin{aligned} \beta_c(s) &= \frac{V_-}{V_o} \Big|_{V_s=0} = \frac{R_1 // Z_{cc}}{R_1 // Z_{cc} + R_2} = \frac{R_1 (1 + sC_c R_c)}{R_2 + sC_c R_2 (R_1 + R_c) + R_1 + sC_c R_1 R_c} = \\ &= \frac{R_1 (1 + sC_c R_c)}{R_1 + R_2 + s(C_c R_2 (R_1 + R_c) + C_c R_1 R_c)} = \frac{R_1}{R_1 + R_2} \cdot \frac{1 + sC_c R_c}{1 + sC_c \frac{R_2 (R_1 + R_c) + R_1 R_c}{R_1 + R_2}} \end{aligned}$$

4) The value of R_c and C_c that yields a $+45^\circ$ phase margin

Hint: place the crossover frequency of loop gain T exactly a decade lower than ω_{p2} and place the compensation network zero at the same frequency.

$$\omega_z = \omega_c = \omega_{p2}/10 = 5 \cdot 10^4 \text{ rad/s}$$

$$A_{OL} \omega_{p1} = \omega_z \stackrel{1}{\beta_c + \infty} \leadsto \frac{1}{\beta_c + \infty} = \frac{A_{OL} \omega_{p1}}{\omega_z} = 200$$

From $\beta_c(s)$ expression:

$$\frac{1}{\beta_c(+\infty)} = \frac{R_1 + R_2}{R_1} \cdot \frac{R_2 (R_1 + R_c) + R_1 R_c}{R_c (R_1 + R_2)} = \frac{R_2 R_1 + R_2 R_c + R_1 R_c}{R_1 R_c} = \frac{R_2 R_1 + R_c (R_2 + R_1)}{R_1 R_c}$$

By equating the two $1/\beta_c + \infty$ expressions:

$$\frac{1}{\beta_c + \infty} R_1 R_c = R_2 R_1 + R_c (R_2 + R_1) \leadsto R_c = \frac{R_2 R_1}{\frac{1}{\beta_c + \infty} R_1 - (R_1 + R_2)} = 285,88 \text{ } \Omega$$

From $\beta_c(s)$ expression:

$$\omega_z = \frac{1}{C_c R_c} = 5 \cdot 10^4 \text{ rad/s} \leadsto C_c = \frac{1}{R_c \omega_z} = 69,86 \text{ nF}$$

Check by calculating ω_p in two ways:

$$\frac{1}{\beta_c + \infty} \cdot \frac{1}{\omega_z} = A_{OL} \cdot \frac{1}{\omega_p} \leadsto \omega_p = \frac{A_{OL} \omega_z}{1/\beta_c + \infty} = 10^3 \text{ rad/s}$$

$$\omega_p = \frac{R_1 + R_2}{C_c (R_2 (R_1 + R_c) + R_1 R_c)} \approx 1020 \text{ rad/s}$$